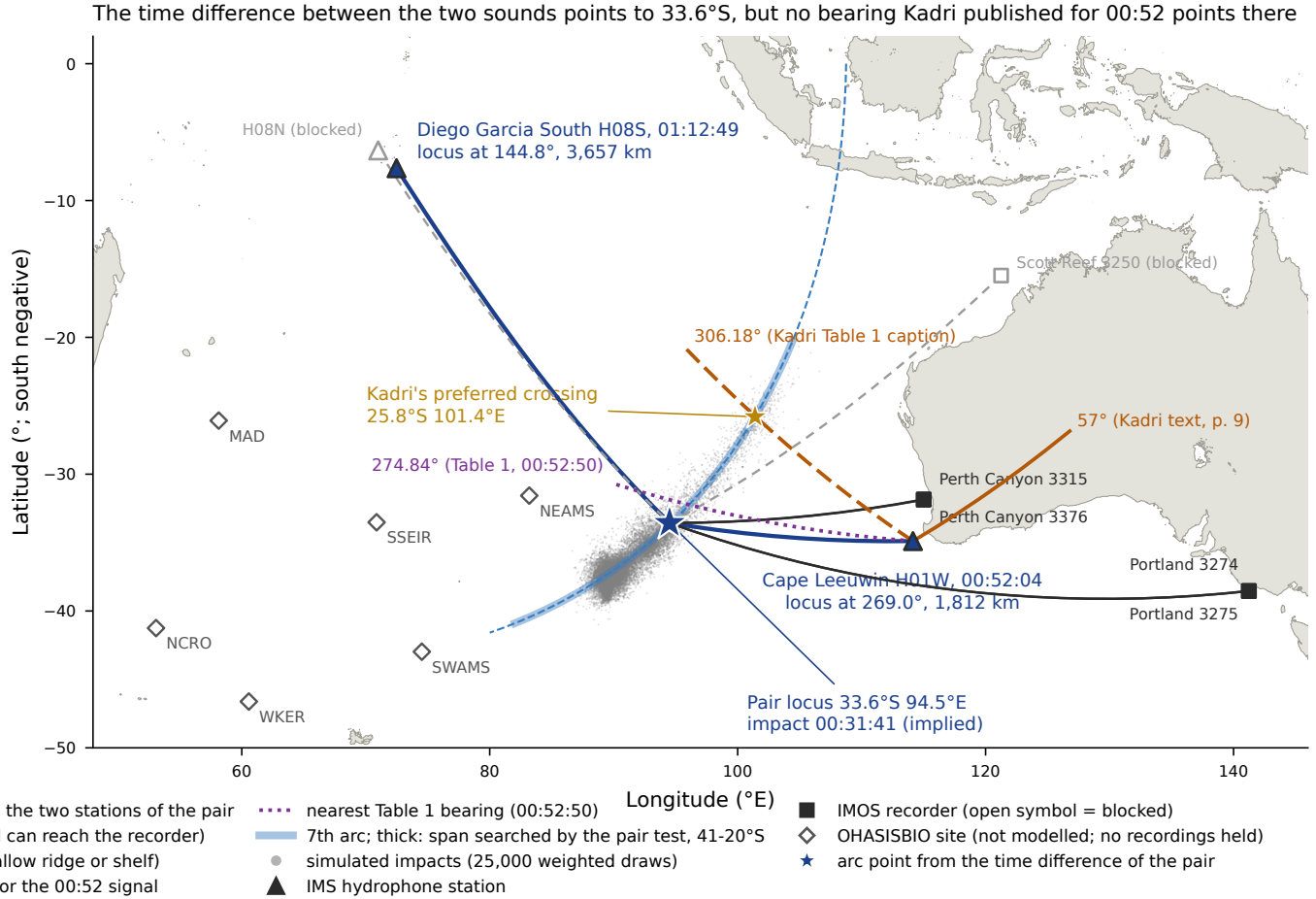


(i) The star is the point on the arc that the time difference of the two sounds gives. The lines show where sound can travel from that point to each recorder. Grey dashed lines are blocked by shallow sea floor. The orange and purple lines are the directions Kadri published from Cape Leeuwin; none goes to the star. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Path status at the 33.5°S node (offset 0, 10 m source): IMS from the KRAKEN arc grid (run-j-tl-grid/tl_grid.csv.gz; H08N no propagating modes, track min depth 6 m); IMOS from the RAM-corrected table imos_tl_ram_arc.csv (3274/3275 open at 130-136 dB for 6.3-40 Hz, KRAKEN gives no modes; 3250 TL > 440 dB, shelf 12 m). OHASISBIO positions from Royer LDO-201608 Table 1. Impact cloud: run J waypoints stratum seeds 1-8 stride 6 + northern supplement (searches geometry/sample.npz, Kish ESS 110k), weights D0'. Bearings are WGS84 geodesics from the H01W triad centroid. Land: GEBCO 2026 at 0.05°. Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.